UCS654

Predictive Analytics

Assignment

Parameter Estimation

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1. $F(x; \theta_1, \theta_2) = \frac{1}{\sqrt{2\pi\theta_2}} \exp\left(\frac{-(x-\theta_1)^2}{2\theta_2}\right)$

$$L(\theta_1, \theta_2) = \frac{1}{\sqrt{2\pi\theta_2}} \exp\left(\frac{-(x_1-\theta_1)^2}{2\theta_2}\right) \cdots \frac{1}{\sqrt{2\pi\theta_2}} \exp\left(\frac{-(x_n-\theta_1)^2}{2\theta_2}\right)$$

$$\Rightarrow L(\theta_1, \theta_2) = (2\pi\theta_2)^{-n/2} \exp\left(\frac{-1}{2\theta_2} \sum (x_i-\theta_1)^2\right)$$

$$\Rightarrow \ln L = \frac{-n}{2} \ln(2\pi\theta_2) - \frac{1}{2\theta_2} \sum (x_i-\theta_1)^2$$

$$\Rightarrow \ln L = \frac{-n}{2} \ln(2\pi) - \frac{n}{2} \ln(\theta_2) - \frac{1}{2\theta_2} \sum (x_i-\theta_1)^2$$

$$\Rightarrow \frac{\partial(\ln L)}{\partial\theta_1} = \frac{1}{\theta_2} \sum (x_i-\theta_1) = 0$$

$$\Rightarrow \sum x_i - n\theta_1 = 0$$

$$\Rightarrow \theta_1 = \frac{\sum x_i}{n}$$

$$\Rightarrow \theta_1 = \bar{x}$$

$$\frac{\partial(\ln L)}{\partial\theta_2} = \frac{-n}{2\theta_2} + \frac{1}{2\theta_2^2} \sum (x_i-\theta_1)^2 = 0$$

$$\Rightarrow \frac{n}{\cancel{2}\cancel{\theta_2}} = \frac{1}{\cancel{2}\theta_2^{\cancel{2}}} \sum (x_i-\theta_1)^2$$

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$\Rightarrow \theta_2 = \frac{1}{n} \sum (x_i - \theta_1)^2$

$\Rightarrow \theta_2 = \sigma^2$

2. $F(x;\theta) = {}^{nm}C_x \, \theta^x (1-\theta)^{nm-x}$

$\pi(\theta) \propto \theta^{\alpha-1} (1-\theta)^{\beta-1}$

$\Rightarrow \pi(\theta|x) \propto \theta^{x+\alpha-1} (1-\theta)^{nm-x+\beta-1}$

$\text{Mean}(\beta) = \frac{a}{a+b}$

$\therefore \hat{\theta}_{MMSE} = \frac{x+\alpha}{(x+\alpha)+(nm-x+\beta)}$

$\Rightarrow \hat{\theta}_{MMSE} = \frac{x+\alpha}{nm+\alpha+\beta}$